

# Many regressions

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## Dependencies

This document depends on the following packages:

```
library(devtools)
library(Biobase)
library(limma)
library(edge)
```

To install these packages you can use the code (or if you are compiling the document, remove the `eval=FALSE` from the chunk.)

```
install.packages(c("devtools"))
source("http://www.bioconductor.org/biocLite.R")
biocLite(c("Biobase", "limma", "jdstorey/edge"))
```

## Download the data

Here we are going to use some data from the paper Detection of redundant fusion transcripts as biomarkers or disease-specific therapeutic targets in breast cancer.

(<http://www.ncbi.nlm.nih.gov/pubmed/22496456>) that uses data from different normal human tissues (called the Illumina BodyMap data).

```

con =url("http://bowtie-bio.sourceforge.net/recount/ExpressionSets/bottomly_eset.RData")
load(file=con)
close(con)
bot = bottomly.eset
pdata=pData(bot)
edata=as.matrix(exprs(bot))
fdata = fData(bot)
ls()

```

```

## [1] "bot"          "bottomly.eset" "con"           "edata"
## [5] "fdata"        "pdata"         "tropical"

```

## Transform the data

Here we will transform the data and remove lowly expressed genes.

```

edata = log2(as.matrix(edata) + 1)
edata = edata[rowMeans(edata) > 10, ]

```

## Fit many regression models at once.

```

mod = model.matrix(~ pdata$strain)
fit = lm.fit(mod,t(edata))
names(fit)

```

```

## [1] "coefficients" "residuals"    "effects"      "rank"
## [5] "fitted.values" "assign"       "qr"          "df.residual"

```

## Compare to output of lm

```
fit$coefficients[,1]
```

```

##      (Intercept) pdata$strainDBA/2J
##      10.4116634      0.3478919

```

```
tidy(lm(as.numeric(edata[1, ]) ~ pdata$strain))
```

```

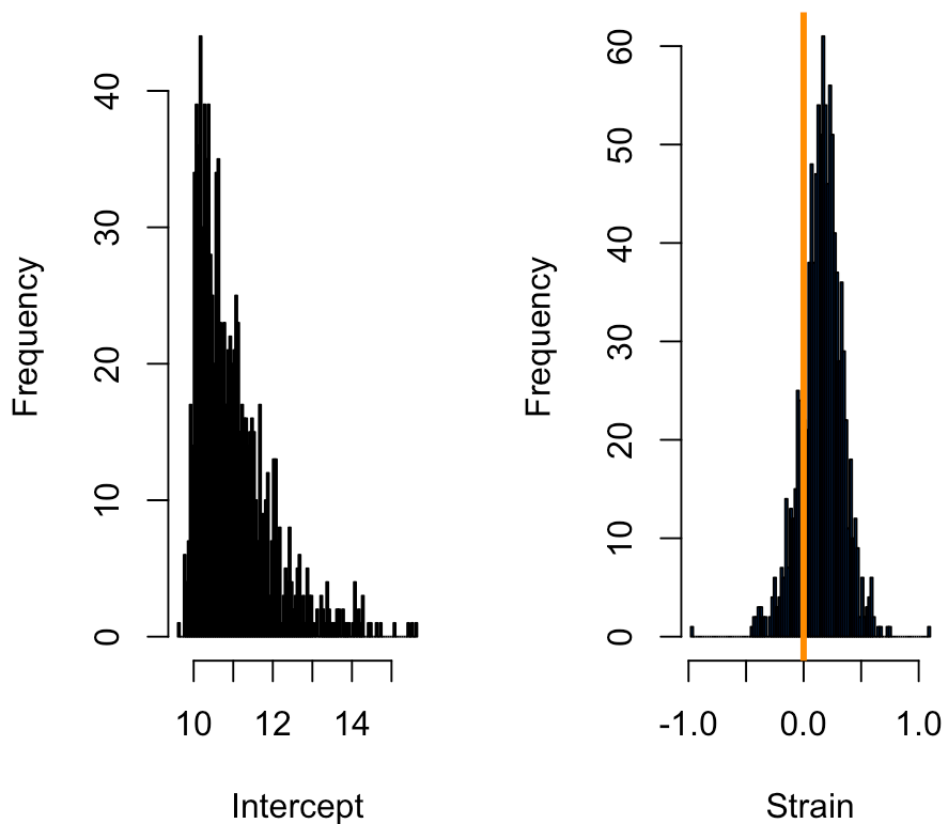
##           term      estimate std.error statistic    p.value
## 1      (Intercept) 10.4116634 0.1516267  68.666422 3.097880e-24
## 2  pdata$strainDBA/2J  0.3478919 0.2095024   1.660563 1.132148e-01

```

## Look at the coefficients across genes

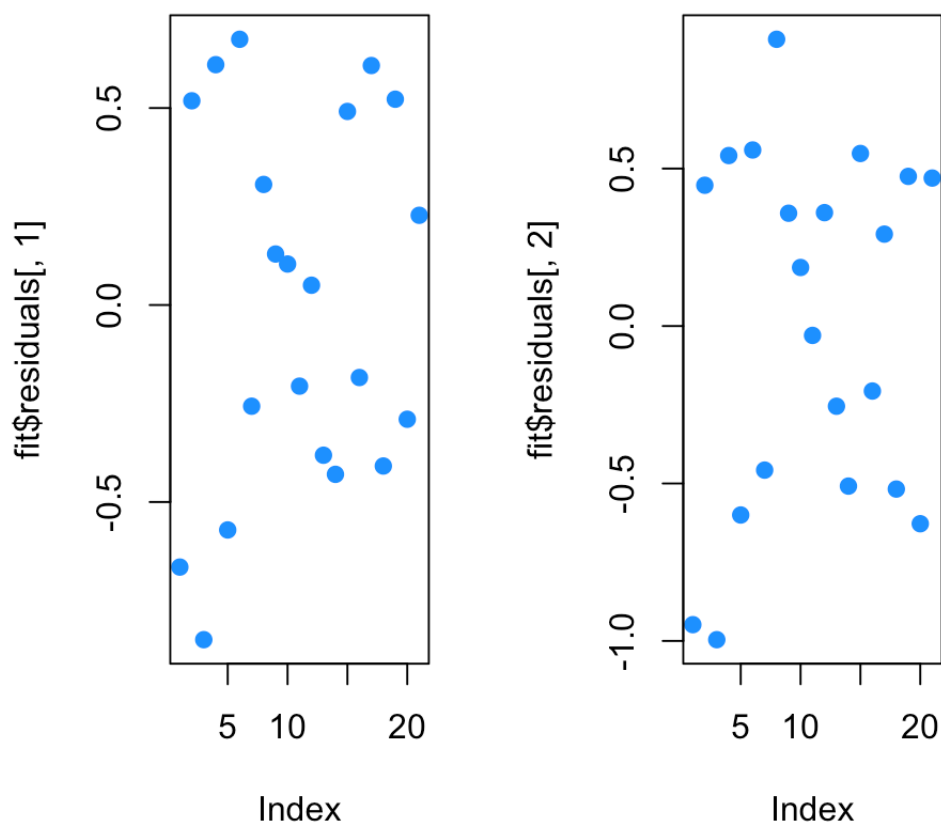
```
par(mfrow=c(1,2))
hist(fit$coefficients[1,],breaks=100,col=2,xlab="Intercept")
hist(fit$coefficients[2,],breaks=100,col=2,xlab="Strain")
abline(v=0,lwd=3,col=1)
```

## histogram of fit\$coefficients histogram of fit\$coefficients



## Look at the residuals for a couple of genes

```
par(mfrow=c(1,2))
plot(fit$residuals[,1],col=2)
plot(fit$residuals[,2],col=2)
```



## Fit many regressions with an adjustment

```
mod_adj = model.matrix(~ pdata$strain + as.factor(pdata$lane.number))
fit_adj = lm.fit(mod_adj,t(edata))
fit_adj$coefficients[,1]
```

```
##              (Intercept)          pdata$strainDBA/2J
##              10.31359781             0.28934825
## as.factor(pdata$lane.number)2 as.factor(pdata$lane.number)3
##              0.05451431             0.02502244
## as.factor(pdata$lane.number)5 as.factor(pdata$lane.number)6
##              0.07200502             0.38038016
## as.factor(pdata$lane.number)7 as.factor(pdata$lane.number)8
##              0.21815863             0.15103858
```

## Fit many regressions with the limma package

```
fit_limma = lmFit(edata,mod_adj)
names(fit_limma)
```

```
## [1] "coefficients"      "rank"                "assign"
## [4] "qr"                 "df.residual"         "sigma"
## [7] "cov.coefficients"   "stdev.unscaled"     "pivot"
## [10] "Amean"              "method"              "design"
```

```
fit_limma$coefficients[1,]
```

```
##                (Intercept)                pdata$strainDBA/2J
##                10.31359781                0.28934825
## as.factor(pdata$lane.number)2 as.factor(pdata$lane.number)3
##                0.05451431                0.02502244
## as.factor(pdata$lane.number)5 as.factor(pdata$lane.number)6
##                0.07200502                0.38038016
## as.factor(pdata$lane.number)7 as.factor(pdata$lane.number)8
##                0.21815863                0.15103858
```

```
fit_adj$coefficients[,1]
```

```
##                (Intercept)                pdata$strainDBA/2J
##                10.31359781                0.28934825
## as.factor(pdata$lane.number)2 as.factor(pdata$lane.number)3
##                0.05451431                0.02502244
## as.factor(pdata$lane.number)5 as.factor(pdata$lane.number)6
##                0.07200502                0.38038016
## as.factor(pdata$lane.number)7 as.factor(pdata$lane.number)8
##                0.21815863                0.15103858
```

## Fit many regressions with the edge package

```
edge_study = build_study(data=edata,grp=pdata$strain,adj.var=as.factor(pdata$lane.number))
fit_edge = fit_models(edge_study)
summary(fit_edge)
```

```

##
## deFit Summary
##
## fit.full:
##          SRX033480 SRX033488 SRX033481 SRX033489 SRX033482
## ENSMUSG00000000131    10.3    10.3    10.4    10.4    10.3
## ENSMUSG00000000149    10.6    10.6    10.8    10.8    10.7
## ENSMUSG00000000374    10.7    10.7    10.7    10.7    10.7
##          SRX033490 SRX033483 SRX033476 SRX033478 SRX033479
## ENSMUSG00000000131    10.3    10.4    10.7    10.5    10.5
## ENSMUSG00000000149    10.7    10.7    11.2    10.9    10.9
## ENSMUSG00000000374    10.7    10.7    11.0    10.8    10.8
##          SRX033472 SRX033473 SRX033474 SRX033475 SRX033491
## ENSMUSG00000000131    10.6    10.7    10.6    10.7    10.7
## ENSMUSG00000000149    10.6    10.7    10.6    10.6    10.6
## ENSMUSG00000000374    10.9    10.9    10.9    10.9    10.9
##          SRX033484 SRX033492 SRX033485 SRX033493 SRX033486
## ENSMUSG00000000131    11.0    11.0    10.8    10.8    10.8
## ENSMUSG00000000149    11.1    11.1    10.9    10.9    10.8
## ENSMUSG00000000374    11.2    11.2    11.0    11.0    10.9
##          SRX033494
## ENSMUSG00000000131    10.8
## ENSMUSG00000000149    10.8
## ENSMUSG00000000374    10.9
##
## fit.null:
##          SRX033480 SRX033488 SRX033481 SRX033489 SRX033482
## ENSMUSG00000000131    10.4    10.4    10.5    10.5    10.4
## ENSMUSG00000000149    10.6    10.6    10.8    10.8    10.7
## ENSMUSG00000000374    10.8    10.8    10.8    10.8    10.7
##          SRX033490 SRX033483 SRX033476 SRX033478 SRX033479
## ENSMUSG00000000131    10.4    10.6    10.9    10.7    10.7
## ENSMUSG00000000149    10.7    10.6    11.1    10.9    10.8
## ENSMUSG00000000374    10.7    10.8    11.1    10.9    10.9
##          SRX033472 SRX033473 SRX033474 SRX033475 SRX033491
## ENSMUSG00000000131    10.4    10.5    10.4    10.6    10.6
## ENSMUSG00000000149    10.6    10.8    10.7    10.6    10.6
## ENSMUSG00000000374    10.8    10.8    10.7    10.8    10.8
##          SRX033484 SRX033492 SRX033485 SRX033493 SRX033486
## ENSMUSG00000000131    10.9    10.9    10.7    10.7    10.7
## ENSMUSG00000000149    11.1    11.1    10.9    10.9    10.8
## ENSMUSG00000000374    11.1    11.1    10.9    10.9    10.9
##          SRX033494
## ENSMUSG00000000131    10.7
## ENSMUSG00000000149    10.8
## ENSMUSG00000000374    10.9
##
## res.full:
##          SRX033480 SRX033488 SRX033481 SRX033489 SRX033482
## ENSMUSG00000000131   -0.567    0.616   -0.806    0.654   -0.498
## ENSMUSG00000000149   -0.793    0.602   -0.986    0.551   -0.523
## ENSMUSG00000000374   -0.491    0.620   -0.697    0.646   -0.427
##          SRX033490 SRX033483 SRX033476 SRX033478 SRX033479

```

```

## ENSMUSG00000000131    0.748    -0.231    0.0237    0.00934    0.0511
## ENSMUSG00000000149    0.636    -0.362    0.5347    0.20909    0.1327
## ENSMUSG00000000374    0.756    -0.232    -0.0977    -0.09356    0.0165
##
## SRX033472 SRX033473 SRX033474 SRX033475 SRX033491
## ENSMUSG00000000131   -0.0493    0.1523    -0.250    -0.345    0.576
## ENSMUSG00000000149    0.1911    0.4355    -0.112    -0.347    0.709
## ENSMUSG00000000374   -0.1285    0.0506    -0.329    -0.455    0.687
##
## SRX033484 SRX033492 SRX033485 SRX033493 SRX033486
## ENSMUSG00000000131   -0.408    0.3841    -0.470    0.461    -0.284
## ENSMUSG00000000149   -0.516   -0.0182    -0.601    0.392    -0.615
## ENSMUSG00000000374   -0.325    0.4227    -0.401    0.495    -0.328
##
## SRX033494
## ENSMUSG00000000131    0.233
## ENSMUSG00000000149    0.482
## ENSMUSG00000000374    0.312
##
## res.null:
##
## SRX033480 SRX033488 SRX033481 SRX033489 SRX033482
## ENSMUSG00000000131   -0.664    0.520    -0.902    0.557    -0.594
## ENSMUSG00000000149   -0.764    0.631    -0.957    0.580    -0.494
## ENSMUSG00000000374   -0.558    0.554    -0.763    0.580    -0.493
##
## SRX033490 SRX033483 SRX033476 SRX033478 SRX033479
## ENSMUSG00000000131    0.651   -0.424    -0.169   -0.184   -0.142
## ENSMUSG00000000149    0.665   -0.303    0.593    0.268    0.191
## ENSMUSG00000000374    0.689   -0.365   -0.231   -0.226   -0.116
##
## SRX033472 SRX033473 SRX033474 SRX033475 SRX033491
## ENSMUSG00000000131    0.1436    0.345   -0.0568   -0.249    0.672
## ENSMUSG00000000149    0.1324    0.377   -0.1709   -0.376    0.680
## ENSMUSG00000000374    0.0044    0.183   -0.1960   -0.389    0.754
##
## SRX033484 SRX033492 SRX033485 SRX033493 SRX033486
## ENSMUSG00000000131   -0.311    0.4805   -0.374    0.557   -0.188
## ENSMUSG00000000149   -0.546   -0.0476   -0.630    0.363   -0.645
## ENSMUSG00000000374   -0.259    0.4892   -0.335    0.561   -0.262
##
## SRX033494
## ENSMUSG00000000131    0.330
## ENSMUSG00000000149    0.453
## ENSMUSG00000000374    0.378
##
## beta.coef:
##
##      [,1]  [,2]  [,3]  [,4]  [,5]  [,6]  [,7]  [,8]
## ENSMUSG00000000131 10.3 0.0545 0.0250 0.0720 0.380 0.218 0.1510 0.2893
## ENSMUSG00000000149 10.6 0.1455 0.0784 0.0596 0.531 0.304 0.2083 -0.0881
## ENSMUSG00000000374 10.7 0.0322 -0.0246 -0.0164 0.273 0.121 0.0576 0.1993
##
## stat.type:
## [1] "lrt"

```

```
fit_edge@beta.coef[1,]
```

```

## [1] 10.31359781 0.05451431 0.02502244 0.07200502 0.38038016 0.21815863
## [7] 0.15103858 0.28934825

```

```
fit_limma$coefficients[1,]
```

```
##                (Intercept)                pdata$strainDBA/2J
##                10.31359781                0.28934825
## as.factor(pdata$lane.number)2 as.factor(pdata$lane.number)3
##                0.05451431                0.02502244
## as.factor(pdata$lane.number)5 as.factor(pdata$lane.number)6
##                0.07200502                0.38038016
## as.factor(pdata$lane.number)7 as.factor(pdata$lane.number)8
##                0.21815863                0.15103858
```

## More information

You can find a lot more information on this model fitting strategy in:

- The limma paper  
(<http://www.bioconductor.org/packages/release/bioc/vignettes/limma/inst/doc/usersguide.pdf>)
- The limma vignette  
(<http://www.bioconductor.org/packages/release/bioc/vignettes/limma/inst/doc/usersguide.pdf>)

## Session information

Here is the session information

```
devtools::session_info()
```



```
## setting value
## version R version 3.2.1 (2015-06-18)
## system x86_64, darwin10.8.0
## ui RStudio (0.99.447)
## language (EN)
## collate en_US.UTF-8
## tz America/New_York
##
## package * version date
## acepack 1.3-3.3 2014-11-24
## annotate 1.46.1 2015-07-11
## AnnotationDbi * 1.30.1 2015-04-26
## assertthat 0.1 2013-12-06
## BiasedUrn * 1.06.1 2013-12-29
## Biobase * 2.28.0 2015-04-17
## BiocGenerics * 0.14.0 2015-04-17
## BiocInstaller * 1.18.4 2015-07-22
## BiocParallel 1.2.20 2015-08-07
## biomaRt 2.24.0 2015-04-17
## Biostrings 2.36.3 2015-08-12
## bitops 1.0-6 2013-08-17
## bladderbatch * 1.6.0 2015-08-26
## broom * 0.3.7 2015-05-06
## caTools 1.17.1 2014-09-10
## cluster 2.0.3 2015-07-21
## colorspace 1.2-6 2015-03-11
## corpcor 1.6.8 2015-07-08
## curl 0.9.2 2015-08-08
## DBI * 0.3.1 2014-09-24
## dendextend * 1.1.0 2015-07-31
## DESeq2 * 1.8.1 2015-05-02
## devtools * 1.8.0 2015-05-09
## digest 0.6.8 2014-12-31
## dplyr * 0.4.3 2015-09-01
## edge * 2.1.0 2015-09-06
## evaluate 0.7.2 2015-08-13
## foreign 0.8-66 2015-08-19
## formatR 1.2 2015-04-21
## Formula * 1.2-1 2015-04-07
## futile.logger 1.4.1 2015-04-20
## futile.options 1.0.0 2010-04-06
## gdata 2.17.0 2015-07-04
## genefilter * 1.50.0 2015-04-17
## geneLenDataBase * 1.4.0 2015-09-06
## geneplotter 1.46.0 2015-04-17
## GenomeInfoDb * 1.4.2 2015-08-15
## GenomicAlignments 1.4.1 2015-04-24
## GenomicFeatures 1.20.2 2015-08-14
## GenomicRanges * 1.20.5 2015-06-09
## genstats * 0.1.02 2015-09-05
## ggplot2 * 1.0.1 2015-03-17
## git2r 0.11.0 2015-08-12
## GO.db 3.1.2 2015-09-06
```

|    |                |               |            |
|----|----------------|---------------|------------|
| ## | goseq          | * 1.20.0      | 2015-04-17 |
| ## | gplots         | * 2.17.0      | 2015-05-02 |
| ## | gridExtra      | 2.0.0         | 2015-07-14 |
| ## | gtable         | 0.1.2         | 2012-12-05 |
| ## | gtools         | 3.5.0         | 2015-05-29 |
| ## | highr          | 0.5           | 2015-04-21 |
| ## | HistData       | * 0.7-5       | 2014-04-26 |
| ## | Hmisc          | * 3.16-0      | 2015-04-30 |
| ## | htmltools      | 0.2.6         | 2014-09-08 |
| ## | httr           | 1.0.0         | 2015-06-25 |
| ## | IRanges        | * 2.2.7       | 2015-08-09 |
| ## | KernSmooth     | 2.23-15       | 2015-06-29 |
| ## | knitr          | * 1.11        | 2015-08-14 |
| ## | lambda.r       | 1.1.7         | 2015-03-20 |
| ## | lattice        | * 0.20-33     | 2015-07-14 |
| ## | latticeExtra   | 0.6-26        | 2013-08-15 |
| ## | lazyeval       | 0.1.10        | 2015-01-02 |
| ## | limma          | * 3.24.15     | 2015-08-06 |
| ## | lme4           | 1.1-9         | 2015-08-20 |
| ## | locfit         | 1.5-9.1       | 2013-04-20 |
| ## | magrittr       | 1.5           | 2014-11-22 |
| ## | MASS           | * 7.3-43      | 2015-07-16 |
| ## | Matrix         | * 1.2-2       | 2015-07-08 |
| ## | MatrixEQTL     | * 2.1.1       | 2015-02-03 |
| ## | memoise        | 0.2.1         | 2014-04-22 |
| ## | mgcv           | * 1.8-7       | 2015-07-23 |
| ## | minqa          | 1.2.4         | 2014-10-09 |
| ## | mnormt         | 1.5-3         | 2015-05-25 |
| ## | munsell        | 0.4.2         | 2013-07-11 |
| ## | nlme           | * 3.1-122     | 2015-08-19 |
| ## | nloptr         | 1.0.4         | 2014-08-04 |
| ## | nnet           | 7.3-10        | 2015-06-29 |
| ## | org.Hs.eg.db   | * 3.1.2       | 2015-07-17 |
| ## | plyr           | 1.8.3         | 2015-06-12 |
| ## | preprocessCore | * 1.30.0      | 2015-04-17 |
| ## | proto          | 0.3-10        | 2012-12-22 |
| ## | psych          | 1.5.6         | 2015-07-08 |
| ## | qvalue         | * 2.0.0       | 2015-04-17 |
| ## | R6             | 2.1.1         | 2015-08-19 |
| ## | RColorBrewer   | 1.1-2         | 2014-12-07 |
| ## | Rcpp           | * 0.12.0      | 2015-07-25 |
| ## | RcppArmadillo  | * 0.5.400.2.0 | 2015-08-17 |
| ## | RCurl          | 1.95-4.7      | 2015-06-30 |
| ## | reshape2       | 1.4.1         | 2014-12-06 |
| ## | rmarkdown      | 0.7           | 2015-06-13 |
| ## | rpart          | 4.1-10        | 2015-06-29 |
| ## | Rsamtools      | 1.20.4        | 2015-06-01 |
| ## | RSkittleBrewer | * 1.1         | 2015-09-05 |
| ## | RSQLite        | * 1.0.0       | 2014-10-25 |
| ## | rstudioapi     | 0.3.1         | 2015-04-07 |
| ## | rtracklayer    | 1.28.9        | 2015-08-19 |
| ## | rversions      | 1.0.2         | 2015-07-13 |
| ## | S4Vectors      | * 0.6.5       | 2015-09-01 |
| ## | scales         | 0.3.0         | 2015-08-25 |

```
## snm 1.16.0 2015-04-17
## snpStats * 1.18.0 2015-04-17
## stringi 0.5-5 2015-06-29
## stringr 1.0.0 2015-04-30
## survival * 2.38-3 2015-07-02
## sva * 3.14.0 2015-04-17
## tidyr 0.2.0 2014-12-05
## UsingR * 2.0-5 2015-08-06
## whisker 0.3-2 2013-04-28
## XML 3.98-1.3 2015-06-30
## xml2 0.1.2 2015-09-01
## xtable 1.7-4 2014-09-12
## XVector 0.8.0 2015-04-17
## yaml 2.1.13 2014-06-12
## zlibbioc 1.14.0 2015-04-17
## source
## CRAN (R 3.2.0)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.0)
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## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## Github (jdstorey/edge@a1947b5)
## CRAN (R 3.2.2)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## Bioconductor
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## Bioconductor
## Bioconductor
## local
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
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## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.1)
## Bioconductor
## Github (alyssafrazee/RSkittleBrewer@0a96a20)
```

```
## CRAN (R 3.2.0)
## CRAN (R 3.2.0)
## Bioconductor
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.2)
## Bioconductor
## Bioconductor
## CRAN (R 3.2.1)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## Bioconductor
## CRAN (R 3.2.0)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## CRAN (R 3.2.1)
## CRAN (R 3.2.2)
## CRAN (R 3.2.0)
## Bioconductor
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It is also useful to compile the time the document was processed. This document was processed on: 2015-09-06.